

## 1. Analisis Set A (Ronde 1-10)

### Data Input:

- $TP = 210$  (Deteksi Benar)
- $FN = 0$  (Gagal Deteksi)
- $FP = 10$  (Salah Alert)
- $TN = 140$  (Normal Benar)
- Total Sampel ( $N$ ) = 360

### Perhitungan:

- **Akurasi (Accuracy):**

$$Accuracy = \frac{210 + 140}{360} = \frac{350}{360} = \mathbf{97.22\%}$$

- **Presisi (Precision):**

$$Precision = \frac{210}{210 + 10} = \frac{210}{220} \times 100\% = \mathbf{95.45\%}$$

- **Recall (Detection Rate):**

$$Recall = \frac{210}{210 + 0} = \frac{210}{210} \times 100\% = \mathbf{100\%}$$

- **F1-Score:**

$$F1 = 2 \times \frac{0.9545 \times 1.0000}{0.9545 + 1.0000} = 2 \times \frac{0.9545}{1.9545} = \mathbf{97.67\%}$$

## 2. Analisis Set B (Ronde 11-20)

### Data Input:

- $TP = 209$  (Deteksi Benar)
- $FN = 1$  (Gagal Deteksi)
- $FP = 20$  (Salah Alert)
- $TN = 140$  (Normal Benar)
- Total Sampel ( $N$ ) = **360**

### Perhitungan:

- **Akurasi (Accuracy):**

$$Accuracy = \frac{209 + 140}{360} = \frac{349}{360} = \mathbf{96.94\%}$$

- **Presisi (Precision):**

$$Precision = \frac{209}{209 + 20} = \frac{209}{229} \times 100\% = \mathbf{91.27\%}$$

- **Recall (Detection Rate):**

$$Recall = \frac{209}{209 + 1} = \frac{209}{210} = \mathbf{99.52\%}$$

- **F1-Score:**

$$F1 = \frac{2 \times 209}{(2 \times 209) + 20 + 1} = \frac{418}{418 + 21} = \frac{410}{439} = \mathbf{95.22\%}$$

### 3. Analisis Set C (Ronde 21-30)

#### Data Input:

- $TP = 209$  (Deteksi Benar)
- $FN = 1$  (Gagal Deteksi)
- $FP = 10$  (Salah Alert)
- $TN = 140$  (Normal Benar)
- Total Sampel ( $N$ ) = **360**

#### Perhitungan:

- **Akurasi (Accuracy):**

$$Accuracy = \frac{209 + 140}{360} = \frac{349}{360} = \mathbf{96.94\%}$$

- **Presisi (Precision):**

$$Precision = \frac{209}{209 + 10} = \frac{209}{219} \times 100\% = \mathbf{95.43\%}$$

- **Recall (Detection Rate):**

$$Recall = \frac{209}{209 + 1} = \frac{209}{210} = \mathbf{99.52\%}$$

- **F1-Score:**

$$F1 = \frac{2 \times 209}{(2 \times 209) + 10 + 1} = \frac{418}{418 + 11} = \frac{410}{429} = \mathbf{97.44\%}$$

### Rekapitulasi Data Total (Ronde 1-30)

Kita menjumlahkan hasil dari ketiga set sebelumnya:

- **Total True Positive (TP):**  $210 + 209 + 209 = 628$
- **Total False Negative (FN):**  $0 + 1 + 1 = 2$
- **Total False Positive (FP):**  $10 + 20 + 10 = 40$
- **Total True Negative (TN):**  $140 + 140 + 140 = 420$

**Total Sampel Data (N):**  $628 + 2 + 40 + 420 = 1.090$  Data Point.

### Perhitungan Metrik Evaluasi & Rumus

#### A. Akurasi (Accuracy)

Mengukur ketepatan sistem secara keseluruhan (deteksi benar + normal benar).

$$Accuracy = \frac{628 + 420}{1.090} \times 100\% = \frac{1048}{1.090} = \mathbf{96.15\%}$$

#### B. Presisi (Precision)

Mengukur kepercayaan terhadap alert yang muncul. Karena tidak ada *False Alarm*, nilai ini sempurna.

$$Precision = \frac{628}{628 + 40} \times 100\% = \mathbf{94.01\%}$$

#### C. Recall (Detection Rate)

Mengukur persentase serangan yang berhasil dideteksi dari total serangan yang dilancarkan.

$$Recall = \frac{628}{628 + 2} \times 100\% = \frac{628}{630} = \mathbf{99.68\%}$$

#### D. F1-Score

Nilai rata-rata harmonik untuk menyeimbangkan Presisi dan Recall.

$$F1 = \frac{2 \times 628}{(2 \times 628) + 40 + 2} = \frac{1256}{1256 + 42} = \frac{1256}{1298} = \mathbf{96.76\%}$$